Closed-Form Coordinate Ascent Variational Inference for Student-t Process Regression with Student-t Likelihood

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TL;DR

We developed an efficient, analytically tractable VI framework for a doubly robust regression, Student-t process regression with Student-t likelihood.

Introduction

Background

Combining a **Student-t processes (TP) prior** with a **Student-t likelihood (TPRT)** yields a doubly robustified regression model, which is effective for modeling heavy-tailedness in real-world data.

Challenges & Approaches

Intractability of TPRT for Conventional coordinate ascent variational inference (CAVI) needs to be addressed.

- **Non-exponential family** nature of the Student-t distribution
  - Employment of the **Gaussian-Gamma scale-mixture** representation of the Student-t distribution
- **Necessary dependency** between the Gaussian and Gamma components of the variational posterior
  - A novel **M-projection-based subroutine** for recoupling

Previous Works

	Approximated Posterior	Inference Type	(Scalability)
Tang et al. (2017)	Gaussian	Deterministic (Laplace Approx.)	No
Xu et al. (2024)	Student-t	Stochastic (MC-based VI)	Yes
Ours	Student-t (Augmented)	Deterministic (Analytic CAVI)	Yes

Model

Student-t Process Regression with Student-t Likelihood (TPRT)

For an observation model:

$$y = f(\mathbf{x}) + \varepsilon,$$

we assume **a TP functional prior**

$f \sim \mathcal{TP}(\nu_f, 0, k)$, i.e. $\mathbf{f} = (f(\mathbf{x}_1), \dots, f(\mathbf{x}_N))^\top$ has a density

$$p(\mathbf{f}) = \mathcal{ST}(\mathbf{f}; \nu_f, \mathbf{0}, \mathbf{K}) = \frac{\Gamma\left(\frac{\nu_f+N}{2}\right)}{\Gamma\left(\frac{\nu_f}{2}\right)(\nu_f \pi)^{N/2}|\mathbf{K}|^{1/2}} \left(1 + \frac{1}{\nu_f}\mathbf{f}^\top \mathbf{K}^{-1}\mathbf{f}\right)^{-\frac{\nu_f+N}{2}},$$

and **i.i.d. Student-t noise**

$$\varepsilon \sim \mathcal{ST}(\nu_\varepsilon, 0, \sigma^2).$$

Data Augmentation

To recast the TPRT model within the exponential family, we augment the original generative model.

Original

$$\mathbf{f} \sim \mathcal{ST}(\mathbf{f}; \nu_f, \mathbf{0}, \mathbf{K}_{XX})$$
$$y_i \mid f_i \sim \mathcal{ST}(y_i; \nu_\varepsilon, f_i, \sigma^2)$$

Augmented

$$r \sim \mathrm{Gam}\left(r; \frac{\nu_f}{2}, \frac{\nu_f}{2}\right)$$
$$\mathbf{f} \mid r \sim \mathcal{N}(\mathbf{f}; \mathbf{0}, r^{-1}\mathbf{K}_{XX})$$
$$\lambda_i \sim \mathrm{Gam}\left(\lambda_i; \frac{\nu_\varepsilon}{2}, \frac{\nu_\varepsilon}{2}\right)$$
$$y_i \mid f_i, \lambda_i \sim \mathcal{N}(y_i; f_i, \lambda_i^{-1}\sigma^2)$$

Inference

Methodology: Closed-Form CAVI for TPRT

Aim: An **"efficient"** construction of **"expressive"** variational posterior

$$\mathcal{ST}(\mathbf{f}) = \int \underbrace{\mathcal{NG}(\mathbf{f}, r)}_{=\mathcal{N}(\mathbf{f};\mathbf{m}, r^{-1}\mathbf{S})\mathrm{Gam}(r;\alpha_r,\beta_r)} dr$$

Primary Objective

$$\underset{\text{Variational}}{\arg\min}\ \mathrm{KL}\left(\underbrace{\mathcal{NG}(\mathbf{f}, r)\prod_{i=1}^{N}\mathrm{Gam}(\lambda_i)}_{\text{Variational}} \,\Big\|\, \underbrace{p(\mathbf{f}, r, \boldsymbol{\lambda} \mid \mathbf{y})}_{\text{True}}\right)$$

Cond. Conj.

Cond. Conj.

Sub Objective

$$\underset{\text{Variational}}{\arg\min}\ \mathrm{KL}\left(\underbrace{\widetilde{\mathcal{N}}(\mathbf{f})\ \widetilde{\mathrm{Gam}(r)}}_{\text{Proxy}} \,\Big\|\, \underbrace{\mathcal{NG}(\mathbf{f}, r)}_{\text{Variational}}\right)$$

Cond. Conj.

M-Proj.

Solution

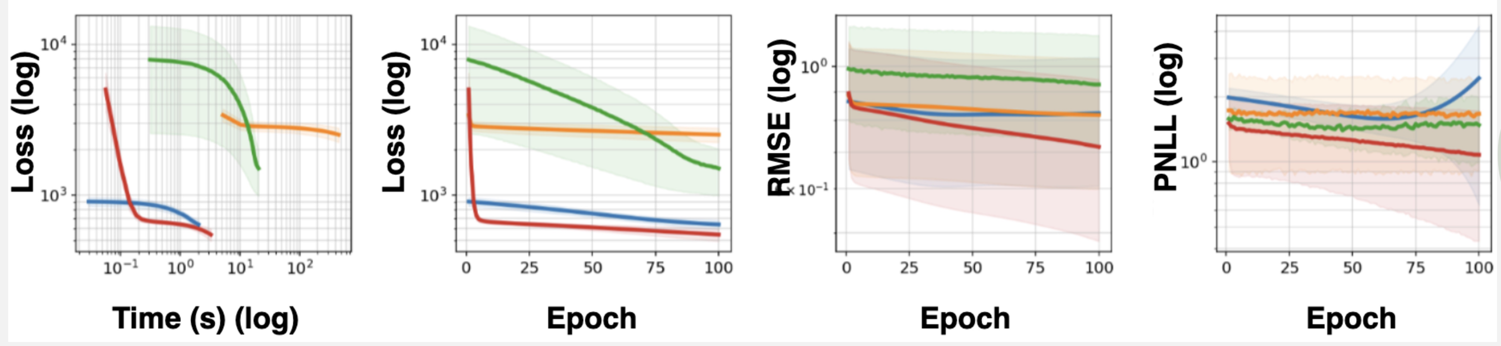
$$\underbrace{\mathcal{NG}^*(\mathbf{f}, r)\prod_{i=1}^{N}\mathrm{Gam}^*(\lambda_i)}_{\text{Optimized variational posterior}}$$

Evaluation

Dense, Full-Batch: 100 Variational EM Iter.

- Our model converged as fast as GPR (No approx).
- Our model demonstrated superior predictive ability.

Boston with outliers GPR, TPR [Tang, 2017], TPR [Xu, 2024], TPR (Ours)



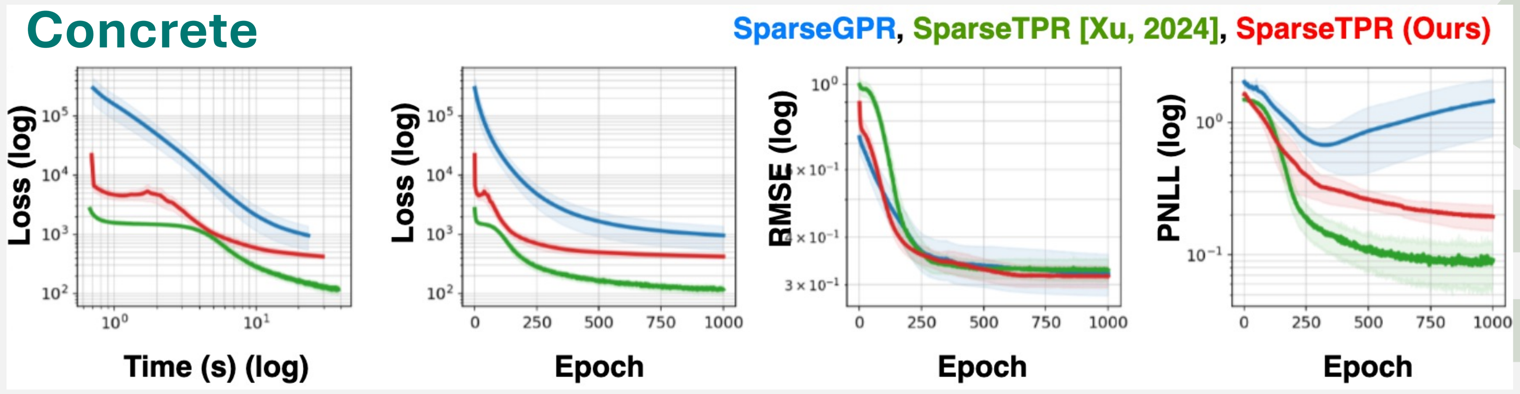
Summary (15 datasets)

Model	Best RMSE	Best PNLL
GPR	3	1
TangTPR	0	0
XuTPR	1	2
TPR (Ours)	**11**	**12**

Sparse, Mini-Batch: 1000 Variational EM Epoch

- Our model "still" demonstrated decent predictive ability.
- Our model suffered from an underestimated variance problem.

Concrete SparseGPR, SparseTPR [Xu, 2024], SparseTPR (Ours)



Summary (14 datasets)

Model	Best RMSE	Best PNLL
SparseGPR	5	1
XuSparseTPR	3	**8**
SparseTPR (Ours)	**6**	5

Discussion on mini-batch settings

- "Shifting goalpost" in V-EM is a potential cause of the performance degradation. (ELBO depends on hyperprams)
- ELBO may not be sufficiently tight.

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Selected References

Tang, Qingtao, et al. "Student-t process regression with student-t likelihood." AAAI, 2017.
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